

CS 6603: AI, Ethics, and Society
Homework Project #5: Fairness & Bias Frequently Asked Questions

Step 1: Dataset Selection

German Dataset

1. **In the German dataset, how do I load the file? What file should I be using?**

In the case of the German Credit dataset, you might want to download the *german.data* file and add the defined columns names for them as specified in its documentation.

This pandas documentation will be useful for the task:

https://pandas.pydata.org/docs/reference/api/pandas.read_table.html

2. **In the German Dataset, what does the 1/2 at the end of each row indicate?**

If the loan was approved, it is set to 1, and if the loan is denied, it is set to 2.

Taiwan Dataset

1. **On the Taiwan Dataset, some of the values don't seem to correspond to the keys given for the X6-X11 columns (-2, 0). Is this a problem?**

See the data documentation: <https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset/discussion/34608>

Portuguese Dataset

1. **For the Portuguese Bank Marketing Data Set, there's a couple of excel files. Are we using one or all of them for this assignment?**

You should choose the best one for completing the project.

2. **For the Portuguese dataset, the "Y" target variable refers to whether a client subscribes to a term deposit (it's not related to loan approval or denial). Is it appropriate to treat this target as a measure of creditworthiness as the instructions allude to?**

Yes.

Step 2: Dataset exploration

1. **Can I consider marital status as a protected class?**

Yes, you may for this assignment.

2. **Should the outcome variable be included in the count for the question "how many variables are in the dataset"?**

No

Step 3.2: Formula to Determine Creditworthiness

1. **If I used an algorithm like Random Forest as my formula, should I just mention that or provide some model equation?**

You can note random forest and include any other description needed to support your equation.

Step 3.5: Split Dataset into Train and Test Sets

1. **Step 3.5 requires the given dataset to be split into equally sized training and test sets (presumably on a 50/50 split), but the rest of the assignment does not seem to indicate the use of the test set. Is this correct?**

Yes, the test set is just used to report on the number of members in the privileged and unprivileged before and after the split. Since you are splitting equally, you can compare your numbers they should be similar.

Step 4: Maximizing Profit

1. **Is this step to be performed on the entire dataset? Or just the training? Or just the testing?**

It is only on the training set.

2. **The profit found is negative. Will I be penalized?**

No, you will not be penalized.

3. **If I determined that threshold 0 will return the maximum profit, do I need to go back change the formula of the Creditworthiness?**

Yes. You should not use 0 as your threshold. In the real world, you would not have a scenario where all loans are approved. To fix this, you can try changing your formula.

If a threshold of 0 produces the best result, then I would evaluate the other threshold results and see where it starts to peak and change direction and use that as your threshold for profit.

4. **For Steps 4 and 6, we are asked to create a table documenting how many in each group (privileged and unprivileged) received Favorable (i.e., Approved) versus Unfavorable (i.e., Declined) outcomes. What exactly should this table look like?**

Here is an example format you could use for the table:

	Privileged <Insert Group>	Unprivileged <Insert Group>
Unfavorable Outcome		
Favorable Outcome		

Step 4.2: Example for Calculating Profit

In this example, we will use the “Generic Loan Approval Dataset.” The outcome variable is Approval_status (0 – Denied; 1 – Approved):

Step 1 Dataset selected: Generic Loan Approval Dataset

Step 3.1 Outcome variable: Approval_status (0 – Denied; 1 - Approved)

Step 3.4 Creditworthiness Formula:

(If Net_Worth > 100K, then 50; else 0) + (If Good_Repayment_History, then 30; else 0)

We will calculate profit for 4 records from the dataset. The values for Net_worth and Good_Repayment_History draw from the Generic Loan Approval Dataset. The Creditworthiness score was determined using the formula in Step 3.4

	Net_Worth	Good_Repayment_History	Creditworthiness Score
Record 1	120K	Yes	80
Record 2	500K	No	50
Record 3	55K	Yes	30
Record 4	10K	No	0

Next, we will walk through variable thresholds and calculate profit.

We will start with a threshold of 1:

Threshold: 0			
	Dataset Approval Status (Outcome variable from Step 3.1)	Creditworthiness Approval Status (Creditworthiness >= X)	Profit
Record 1	Approved	Approved	+10
Record 2	Approved	Approved	+10
Record 3	Denied	Approved	-10
Record 4	Denied	Approved	-10
<i>Total:</i>			0

Next, we will increase the threshold to 20:

Threshold: 20			
	Dataset Approval Status (Outcome variable from Step 3.1)	Creditworthiness Approval Status (Creditworthiness >= X)	Profit
Record 1	Approved	Approved	+10
Record 2	Approved	Approved	+10
Record 3	Denied	Approved	-10
Record 4	Denied	Denied	0
<i>Total:</i>			10

Increasing the threshold to 40:

Threshold: 40			
	Dataset Approval Status (Outcome variable from Step 3.1)	Creditworthiness Approval Status (Creditworthiness >= X)	Profit
Record 1	Approved	Approved	+10
Record 2	Approved	Approved	+10
Record 3	Denied	Denied	0
Record 4	Denied	Denied	0
<i>Total:</i>			+20

Increasing the threshold to 60:

Threshold: 60			
	Dataset Approval Status (Outcome variable from Step 3.1)	Creditworthiness Approval Status (Creditworthiness $\geq X$)	Profit
Record 1	Approved	Approved	+10
Record 2	Approved	Denied	-5
Record 3	Denied	Denied	0
Record 4	Denied	Denied	0
<i>Total:</i>			+5

Increasing the threshold to 90:

Threshold: 90			
	Dataset Approval Status (Outcome variable from Step 3.1)	Creditworthiness Approval Status (Creditworthiness $\geq X$)	Profit
Record 1	Approved	Denied	-5
Record 2	Approved	Denied	-5
Record 3	Denied	Denied	0
Record 4	Denied	Denied	0
<i>Total:</i>			-10

For this example, a threshold of $X=40$ resulted in a max profit of +20. Any threshold $31 \leq X \leq 50$ would also maximize profit.

Step 5: Compute Fairness Metrics on Training Set

1. In Step 5 (and technically Step 6), is it ok to calculate the fairness metrics without using any open-source tool kit?

Yes, if you are having difficulties with the open-source tool kit, you can calculate the metrics by writing your own code.

Step 6: Mitigate Bias in Dataset

1. Will there be a penalty if the profit is zero or negative?

No, see step 4 of FAQ above.

2. How do I make the table for this step?

See step 4 of the FAQ above.

Step 8: Submission

1. Do we need to save and submit the test and training sets as a file or part of the PDF?

No, the train and test sets are not needed for submission.

2. Do we need to submit code files?

No, only the report (PDF) is required for submission.